



Graphing Exponential Functions

Algebra 2 Worksheet · Grade 9-11

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Learning Objectives

- Identify the y-intercept of an exponential function from its equation
- Determine whether an exponential function is increasing or decreasing
- Sketch the graph of an exponential function with a leading coefficient

For each exponential function, identify the y-intercept, state whether the graph is increasing or decreasing, and sketch the graph showing the horizontal asymptote.

1. Identify the y-intercept and behavior of the function.

$$y = 2^x$$

Answer: _____

2. Identify the y-intercept and behavior of the function.

$$y = 3^{-x}$$

Answer: _____

3. Identify the y-intercept and behavior of the function.

$$y = \frac{1}{2} \cdot 5^x$$

Answer: _____

4. Identify the y-intercept and behavior of the function.

$$y = \frac{1}{3} \cdot 4^x$$

Answer: _____

5. Identify the y-intercept and behavior of the function.

$$y = 5 \cdot \left(\frac{1}{2}\right)^x$$

Answer: _____

6. Identify the y-intercept and behavior of the function.

$$y = \left(\frac{1}{4}\right)^x$$

Answer: _____



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7. Identify the y-intercept and behavior of the function.

$$y = \frac{1}{2^x}$$

Answer: _____

8. Identify the y-intercept and behavior of the function.

$$y = 3 \cdot 2^x$$

Answer: _____

9. Identify the y-intercept and behavior of the function.

$$y = 4 \cdot 3^{-x}$$

Answer: _____

10. Identify the horizontal asymptote of every exponential function in this worksheet.

$$y = a \cdot b^x$$

Answer: _____





Remind students that the horizontal asymptote for all functions below is $y = 0$ and that the leading coefficient determines the y -intercept.

Solutions

1. Identify the y -intercept and behavior of the function.

$$y = 2^x$$

- Substitute x equals zero into the function.
- Two raised to the zero power equals one, so the y -intercept is at $(0, 1)$.
- Since the base 2 is greater than 1 and the exponent is positive, the graph is increasing.

Answer: y -intercept: $(0, 1)$; increasing

2. Identify the y -intercept and behavior of the function.

$$y = 3^{-x}$$

- Substitute x equals zero into the function.
- Three raised to the zero power equals one, so the y -intercept is at $(0, 1)$.
- Because the exponent is negative, the graph is decreasing from left to right.

Answer: y -intercept: $(0, 1)$; decreasing

3. Identify the y -intercept and behavior of the function.

$$y = \frac{1}{2} \cdot 5^x$$

- The leading coefficient one-half determines the y -intercept.
- When x equals zero, the function value is one-half, so the y -intercept is $(0, 1/2)$.
- Since the base 5 is greater than 1 and the exponent is positive, the graph is increasing.

Answer: y -intercept: $(0, \frac{1}{2})$; increasing

4. Identify the y -intercept and behavior of the function.

$$y = \frac{1}{3} \cdot 4^x$$

- The leading coefficient one-third gives the y -intercept.
- When x equals zero, the function value is one-third, so the y -intercept is $(0, 1/3)$.
- Since the base 4 is greater than 1 and the exponent is positive, the graph is increasing.

Answer: y -intercept: $(0, \frac{1}{3})$; increasing



5. Identify the y-intercept and behavior of the function.

$$y = 5 \cdot \left(\frac{1}{2}\right)^x$$

→ The leading coefficient 5 determines the y-intercept.

→ When x equals zero, the function value is 5, so the y-intercept is (0, 5).

→ Because the base one-half is less than 1, the graph is decreasing from left to right.

Answer: y-intercept : (0, 5); decreasing

6. Identify the y-intercept and behavior of the function.

$$y = \left(\frac{1}{4}\right)^x$$

→ Substitute x equals zero into the function.

→ Any nonzero base raised to the zero power equals one, so the y-intercept is (0, 1).

→ Since the base one-fourth is less than 1, the graph is decreasing.

Answer: y-intercept : (0, 1); decreasing

7. Identify the y-intercept and behavior of the function.

$$y = \frac{1}{2^x}$$

→ Rewrite the function as 2 raised to the negative x power.

→ When x equals zero, the function value is one, so the y-intercept is (0, 1).

→ Because the rewritten exponent is negative, the graph is decreasing.

Answer: y-intercept : (0, 1); decreasing

8. Identify the y-intercept and behavior of the function.

$$y = 3 \cdot 2^x$$

→ The leading coefficient 3 gives the y-intercept.

→ When x equals zero, the function value is 3, so the y-intercept is (0, 3).

→ Since the base 2 is greater than 1 and the exponent is positive, the graph is increasing.

Answer: y-intercept : (0, 3); increasing

9. Identify the y-intercept and behavior of the function.

$$y = 4 \cdot 3^{-x}$$

→ The leading coefficient 4 determines the y-intercept.

→ When x equals zero, three to the zero power equals one, so the function value is 4 and the y-intercept is (0, 4).

→ Because the exponent is negative, the graph is decreasing from left to right.

Answer: y-intercept : (0, 4); decreasing

10. Identify the horizontal asymptote of every exponential function in this worksheet.

$$y = a \cdot b^x$$

→ All basic exponential functions of this form approach but never touch the x -axis.

→ Therefore the horizontal asymptote is the line y equals zero.

Answer: $y = 0$

